

HW 2 – Linear Algebra

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Solutions:

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Part A: Linear Combination

$$1- \quad u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad v = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad : \quad w = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$a\vec{u} + b\vec{v} = \vec{w}$$

$$a + 2b = 5 \quad \dots i$$

$$2a + b = 4 \quad \dots ii$$

$$\left[\begin{array}{cc|c} a & 2b & 5 \\ 2a & b & 4 \end{array} \right] \quad (\text{Augmented Matrix form})$$

$$2R_1 \rightarrow R_2 = R_2'$$

$$\left[\begin{array}{cc|c} a & 2b & 5 \\ 0 & 3b & 6 \end{array} \right]$$

$$3b = 6 \Rightarrow b = 2$$

Put $b=2$ in eq R_1

$$a + 2(2) = 5$$

$$a = 5 - 4 \Rightarrow a = 1$$

2- 'a' and 'b' are number of units for the movement of objects in the game. If it's negative, the movement is negative.

Part B:- Span

3-

$$a\vec{u} + b\vec{v} = \vec{w}$$

The general movement can be produced by defining the direction of movement via $\vec{u}$ and $\vec{v}$ vectors.

Since 'a' and 'b' are the scalars, they will increase or decrease the range of vectors.

4- Spanning $\vec{u}$, and $\vec{v}$ means to increase the accessible regions and movements in the game. 'a' and 'b' are scalars which can be any positive or negative number so we can make any movements but in the direction of $\vec{u}$ and $\vec{v}$.

5-

$$a\vec{u} + b\vec{v} = \vec{q}$$

$$a\begin{bmatrix} 1 \\ 2 \end{bmatrix} + b\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$$

$$a + 2b = 7 \quad \text{--- (i)}$$

$$2a + b = 5 \quad \text{--- (ii)}$$

$$\left[\begin{array}{cc|c} a & 2b & 7 \\ 2a & b & 5 \end{array} \right] \quad (\text{Augmented Matrix Form})$$

$$R_2' = 2R_1 - R_2$$

$$\left[\begin{array}{cc|c} a & 2b & 7 \\ 0 & 3b & 9 \end{array} \right]$$

$$3b = 9 \Rightarrow b = 3$$

Put $b = 3$ in R_1

$$a + 2(3) = 7$$

$$a = 7 - 6 \Rightarrow a = 1$$

The target $\vec{q}$ is achievable by spanning the vector $\vec{v}$ by $b = 3$.

Part C: Homogenous System.

6.

$$a\vec{u} + b\vec{v} = \vec{0}$$

$$a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \vec{0}$$

$$a + 2b = 0 \quad \dots (i)$$

$$2a + b = 0 \quad \dots (ii)$$

7-

$$\begin{bmatrix} a & 2b & : & 0 \\ 2a & b & : & 0 \end{bmatrix} \text{ (Augmented Matrix)}$$

$$R_2' = 2R_1 - R_2$$

$$\begin{bmatrix} a & 2b & : & 0 \\ 0 & 3b & : & 0 \end{bmatrix}$$

$$3b = 0 \Rightarrow b = 0$$

Put $b=0$ in R₁

$$a + 2(a) = 0$$

$$a + 2a = 0 \Rightarrow a = 0$$

8- $a = b = 0$, hence as a homogeneous equation, it has a trivial solution. The target is $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ so zero movement is required or movement in a way that it cancels out and keeps the object at origin if we provide a non-zero solution.

Part D: Matrix Representation:-

$$9- a\vec{u} + b\vec{v} = \vec{w} ; A\vec{x} = \vec{w}$$

$$\vec{x} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} a \\ b \end{bmatrix}, \vec{w} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$\text{So } A\vec{x} = \vec{w} \Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

10- Augmented Matrix Form

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & 1 & 4 \end{array} \right]$$

$$R_2' = 2R_1 - R_2$$

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 3 & 6 \end{array} \right]$$

$$3b = 6 \Rightarrow \boxed{b = 2}$$

Put $b = 2$ in R_1

$$a + 2b = 5$$

$$a = 5 - 4 \Rightarrow \boxed{a = 1}$$

Challenge Question:

$$P = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, Q = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, u = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, v = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\vec{x} = \vec{p} + s\vec{u} \quad \&$$

$$\vec{x} = \vec{q} + t\vec{v}$$

$$\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} + s \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$-1 + s = 2 - t \quad \dots i$$

$$2 + 2s = 2 + t \quad \dots ii$$

$$1 + s = 0 \quad \dots iii$$

$$\boxed{s = -1}$$

$$2 - 2 = 2 + t$$

$$\boxed{t = -2}$$

Put $s = -1$ & $t = -2$ in eq (i)

$$-1 - 1 = 2 + 2$$

$-2 = 4$ Inconsistent system.